

SISSETON, SD, USA

WMO: 726519

Lat: 45.669N

Lon: 96.991W

Elev: 354

StdP: 97.15

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 94993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.9	-29.1	0.3	-24.6	-26.9	0.3	-22.8	15.7	-7.3	14.5	-6.7	4.1	320	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.3	22.5	30.7	21.9	29.0	20.9	24.7	30.1	23.6	28.5	22.5	27.3	6.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.5	27.7	22.0	17.4	27.0	20.9	16.3	25.8	77.0	30.2	72.3	28.4	68.0	27.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.9	10.8	DB	-30.0	35.4	2.9	1.8	-32.1	36.7	-33.8	37.8	-35.5	38.8	-37.6	40.1	(4)
(5)			WB	-30.1	26.6	2.7	1.1	-32.0	27.4	-33.6	28.0	-35.1	28.6	-37.0	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.8	-9.2	-1.1	6.7	13.8	19.6	22.1	20.2	16.3	8.3	0.6	-6.6	(6)	
	DBStd	12.70	7.99	7.23	7.39	5.88	5.15	3.42	3.14	3.23	4.75	5.24	6.28	7.46	(7)	
	HDD10.0	2551	614	539	349	127	22	0	0	0	5	96	284	515	(8)	
	HDD18.3	4513	873	772	603	348	158	25	6	17	93	314	531	772	(9)	
	CDD10.0	1395	0	0	5	28	139	288	374	317	194	44	4	1	(10)	
	CDD18.3	316	0	0	0	1	18	63	122	76	33	3	0	0	(11)	
	CDH23.3	3253	0	0	2	25	246	628	1218	721	367	46	0	0	(12)	
	CDH26.7	1045	0	0	0	6	83	189	426	213	116	13	0	0	(13)	
(14) Wind	WSAvg	4.8	5.4	5.3	5.3	5.5	5.4	4.4	3.7	3.5	4.2	4.9	5.1	5.3	(14)	
(15) Precipitation	PrecAvg	649	19	20	33	62	76	113	90	67	71	60	26	19	(15)	
	PrecMax	768	53	44	98	131	139	208	185	115	177	180	98	51	(16)	
	PrecMin	503	0	3	0	5	31	43	25	15	3	8	0	5	(17)	
	PrecStd	89	14	12	21	31	29	46	40	31	41	47	26	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	11.5	21.3	26.3	32.4	33.2	35.2	33.1	32.8	28.5	19.1	15.9	(19)	
		MCWB	3.4	8.1	13.2	14.6	19.4	22.0	24.7	24.0	22.1	17.4	11.7	4.7	(20)	
	2%	DB	5.4	6.7	16.7	22.5	29.0	30.7	32.3	31.1	29.9	23.4	15.6	8.9	(21)	
		MCWB	1.9	3.4	10.8	12.5	17.7	21.1	24.0	22.7	20.9	14.6	9.5	3.6	(22)	
	5%	DB	3.4	3.3	12.6	19.5	26.1	28.8	30.6	28.9	27.5	20.0	12.4	5.6	(23)	
		MCWB	1.0	1.3	8.2	11.6	16.2	19.7	22.9	21.7	19.2	13.1	7.4	2.0	(24)	
	10%	DB	1.4	1.3	8.9	16.9	23.0	27.1	29.0	27.4	25.2	17.2	9.3	2.9	(25)	
		MCWB	-0.1	-0.2	5.9	10.0	14.9	19.4	21.9	20.7	18.2	11.4	5.3	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	7.7	15.1	16.0	21.2	24.9	26.7	25.2	23.7	18.1	12.8	6.2	(27)	
		MCDB	7.8	12.0	18.4	23.5	29.3	29.8	31.3	30.8	29.5	25.2	18.0	11.4	(28)	
	2%	WB	2.2	3.5	11.3	13.8	19.2	23.2	25.2	24.0	22.0	15.8	10.1	3.6	(29)	
		MCDB	4.7	6.2	15.8	20.9	25.7	28.2	30.7	29.1	27.8	22.0	14.7	8.5	(30)	
	5%	WB	1.2	1.5	8.4	12.0	17.8	21.6	24.1	22.8	20.6	13.8	7.7	2.0	(31)	
		MCDB	3.2	3.3	12.5	17.7	23.5	26.8	29.2	27.4	25.7	19.2	12.0	5.2	(32)	
	10%	WB	0.0	-0.2	5.7	10.5	16.1	20.3	22.9	21.6	19.0	11.8	5.6	0.6	(33)	
		MCDB	1.7	1.3	8.9	15.9	21.6	25.2	27.5	26.1	23.8	16.3	8.8	2.9	(34)	
(35) Mean Daily Temperature Range	MDBR		9.3	9.5	10.0	11.9	12.3	11.3	11.9	12.1	12.8	11.0	9.9	8.5	(35)	
	5% DB	MCDBR	11.1	11.0	15.7	17.6	16.7	14.2	13.4	13.7	15.3	16.6	14.3	11.5	(36)	
		MCWBR	8.8	8.6	10.1	9.6	8.1	6.9	6.8	6.9	7.4	9.2	9.0	8.0	(37)	
	5% WB	MCDBR	10.4	10.8	15.0	15.3	14.1	11.9	12.1	12.4	13.4	14.6	14.0	11.0	(38)	
		MCWBR	8.4	8.6	10.0	9.0	7.7	7.0	6.8	7.0	7.5	8.6	9.2	8.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.277	0.288	0.310	0.356	0.377	0.388	0.388	0.395	0.359	0.322	0.287	0.266	(40)	
	taud		2.373	2.371	2.425	2.333	2.354	2.369	2.366	2.332	2.415	2.490	2.493	2.426	(41)	
	Ebn at Noon		835	897	925	902	890	879	874	853	858	848	823	814	(42)	
	Edh at Noon		73	89	98	118	120	119	118	117	99	79	65	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.74	2.83	3.82	4.86	5.53	6.21	6.54	5.48	4.21	2.66	1.72	1.27	(44)	
	RadStd		0.18	0.30	0.34	0.36	0.38	0.36	0.26	0.47	0.28	0.35	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page